CHAPTER 11

LAND RESOURCES AND AGRICULTURE

CHAPTER ESSENCE AT A GLANCE:
Examine land-use classification by revenue records, cropping seasons (Kharif, Rabi, Zaid), major food grains (rice, wheat, coarse cereals, pulses), cash crops (cotton, jute, sugarcane, tea, coffee), and key challenges facing Indian agriculture.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Land-Use Classification Categories:** Revenue records categorize land into nine classes: Forests, Non-Agricultural Uses, Barren Wasteland, Grazing Land, Tree Crops, Culturable Wasteland (fallow 5+ years), Fallow other than Current Fallow (1 to 5 years), Current Fallow (1 year or less), and Net Sown Area (NSA).

• **Three Indian Cropping Seasons:** Kharif: Coincides with the southwest monsoon (June to September; rice, cotton, jowar, bajra). Rabi: Winter season (October to March; wheat, gram, mustard). Zaid: Short summer season (April to June; vegetables, melons, fodder).

• **Major Food Grains and Fibres:** Rice is the primary staple (Aus, Aman, and Boro grown in West Bengal). Wheat is the premier temperate winter crop (UP, Punjab, Haryana). Cotton is cultivated on semi-arid black regur soils, and Jute in the Ganga delta.

• **Problems of Indian Agriculture:** Indian agriculture suffers from erratic monsoon dependence, low per-hectare productivity, extreme land fragmentation, high indebtedness, soil degradation, and lack of modern cold-storage infrastructure.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Cropping Intensity } (\%) = \frac{\text{Gross Cropped Area (GCA)}}{\text{Net Sown Area (NSA)}} \times 100$$

Agricultural Cropping Intensity Index

$$\text{Kharif Season} \iff \text{South-West Monsoon} \rightarrow \{\text{Rice, Cotton, Jowar, Bajra, Jute}\}$$

Monsoon Cropping Season Portfolio

$$\text{Rabi Season} \iff \text{Winter Sowing} \rightarrow \{\text{Wheat, Gram, Mustard, Barley}\}$$

Winter Cropping Season Portfolio

$$\text{Agricultural Productivity} = \frac{\text{Total Agricultural Production Yield (kg)}}{\text{Cultivated Land Area (Hectares)}}$$

Yield Productivity per Cultivated Hectare

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Net Sown Area represents the physical extent of land cultivated during the year, whereas Gross Cropped Area counts every sown plot as many times as it is cropped in a year.

• **Alert:** Culturable Wasteland is arable land left fallow for more than five years, whereas land uncultivated for one to five years is designated as 'fallow other than current fallow'.

• **Alert:** In West Bengal, farmers cultivate three successive crops of paddy in a single agricultural year named Aus (autumn), Aman (winter), and Boro (summer).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Net Sown Area and Gross Cropped Area. How is cropping intensity calculated? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Net Sown Area (NSA):** The physical area of land on which crops are sown and harvested at least once in an agricultural year. **[1 Mark]**

• **Gross Cropped Area (GCA):** The total area sown once as well as more than once in an agricultural year (counting multiple harvests on the same plot). **[1 Mark]**

• **Cropping Intensity:** Expressed as a percentage: Cropping Intensity = (Gross Cropped Area divided by Net Sown Area) multiplied by 100. **[1 Mark]**

2. Differentiate between current fallow and culturable wasteland. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Current Fallow:** Arable cropland left uncultivated for one year or less to naturally replenish soil fertility through natural resting. **[1.5 Marks]**

• **Culturable Wasteland:** Arable land that has been left uncultivated for more than five consecutive agricultural years; it can be rehabilitated through reclamation. **[1.5 Marks]**

3. Explain the three cropping seasons in India with their sowing periods and two representative crops each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Kharif:** Sown in June and harvested in September and October (southwest monsoon period); crops include rice, cotton, and jowar. **[1 Mark]**

• **2. Rabi:** Sown in October and November and harvested in March and April (winter season); crops include wheat, gram, and mustard. **[1 Mark]**

• **3. Zaid:** Short summer season sown in April and harvested in June; crops include watermelon, cucumber, and leafy fodder. **[1 Mark]**

4. Compare the geographical requirements, growing seasons, and leading producing states of rice and wheat in India. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Rice Requirements:** Tropical kharif crop requiring high temperatures (above 25 degrees Celsius), high humidity, and annual rainfall exceeding 100 cm; leading states include West Bengal, Uttar Pradesh, and Punjab. **[2 Marks]**

• **Wheat Requirements:** Temperate rabi crop requiring a cool growing season (10 to 15 degrees Celsius), bright sunny weather during ripening, and 50 to 75 cm rainfall; leading states include Uttar Pradesh, Punjab, and Madhya Pradesh. **[2 Marks]**

5. What are the names of the three rice crops grown in West Bengal in an agricultural year? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Aus:** Autumn paddy crop sown in May and June and harvested in September and October. **[1 Mark]**

• **Aman:** Winter paddy crop (the most important harvest) sown in June and July and harvested in November and December. **[1 Mark]**

• **Boro:** Summer paddy crop cultivated under canal or tubewell irrigation, sown in November and harvested in March and April. **[1 Mark]**

6. Distinguish between dryland farming and wetland farming in India. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Dryland Farming:** Confined to semi-arid regions with annual rainfall less than 75 cm; crops are drought-hardy (bajra, ragi, pulses) with emphasis on soil moisture conservation. **[1.5 Marks]**

• **Wetland Farming:** Practiced in humid regions receiving plentiful rainfall or dependable canal irrigation; crops have high water demand (rice, jute, sugarcane) with flood control measures. **[1.5 Marks]**

7. Discuss any four major problems confronting Indian agriculture and suggest corrective measures. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Erratic Monsoon Dependence:** High vulnerability of rainfed crops to drought or untimely floods. **[1 Mark]**

• **2. Sub-division and Fragmentation of Holdings:** Small average farm size hinders mechanization and leads to uneconomic cultivation. **[1 Mark]**

• **3. Soil Degradation:** Excessive canal flood-irrigation and fertilizer overuse cause salinization and loss of organic matter. **[1 Mark]**

• **4. Corrective Policy Measures:** Micro-irrigation (drip and sprinkler), land consolidation, institutional credit access, and comprehensive cold chain logistics. **[1 Mark]**